



Configuration d'un VPN client vers un site avec OpenVPN

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Introduction

OpenVPN est un logiciel libre permettant de créer un réseau privé virtuel (VPN). Son développement a commencé le 13 mai 2001 grâce à James Yonan.

OpenVPN permet à des pairs de s'authentifier entre eux à l'aide d'une clé privée partagée à l'avance, de certificats électroniques ou de couples de noms d'utilisateur/mot de passe. Il utilise de manière intensive la bibliothèque d'authentification OpenSSL ainsi que le protocole SSLv3/TLSv1. Disponible avec une multitude d'environnements tel que Solaris, OpenBSD, FreeBSD, NetBSD, Linux (Debian, Redhat, Ubuntu, etc.), Mac OS X, Windows 2000, XP, Vista, 7, 8 et 10, il offre de nombreuses fonctions de sécurité et de contrôle.

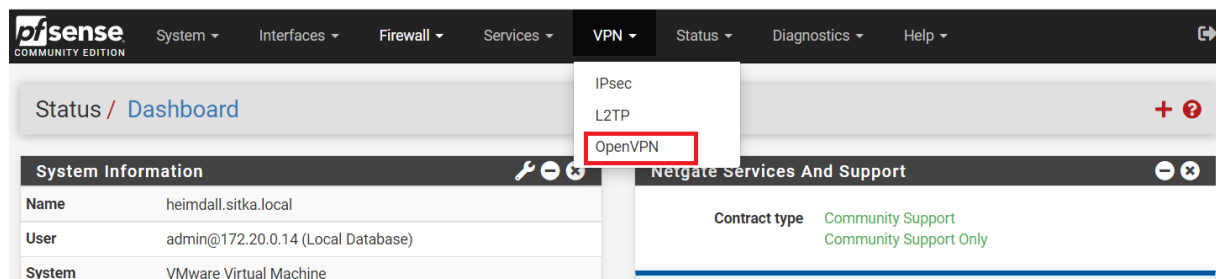
OpenVPN n'est pas compatible avec IPsec ou d'autres logiciels VPN. Le logiciel contient un exécutable pour les connexions du client et du serveur, un fichier de configuration optionnel et une ou plusieurs clés suivant la méthode d'authentification choisie.

1- Gestion des certificats

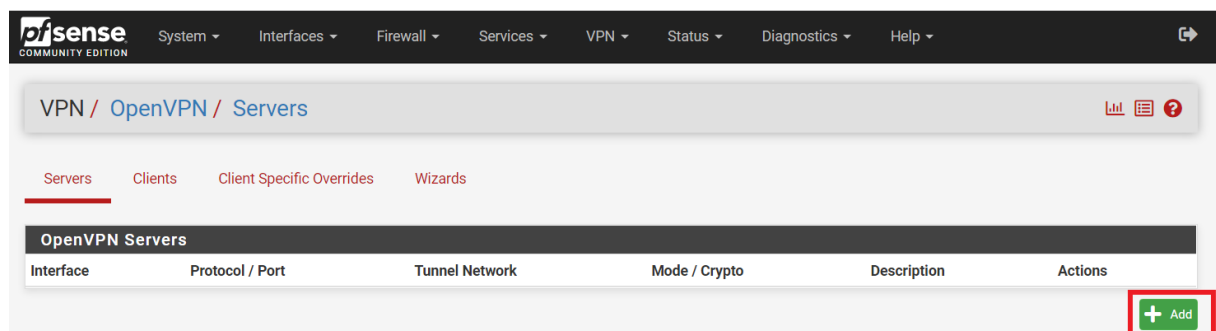
- Création de l'autorité de certification
- Création du certificat Server

2- Configurer le serveur OpenVPN

Pour configurer notre VPN on va cliquer sur le menu "VPN" puis "OpenVPN"



Dans l'onglet **Servers** on clique sur **Add** afin de créer une nouvelle configuration.



Dans "Server Mode" on choisit : **Remote Access (User Auth)**.

Le VPN utilise UDP, avec le port 1194 par défaut on choisira le port 1919 pour des raisons de sécurité.

L'interface, utilisée est la WAN c'est l'interface qui sera utilisé pour notre connexion à distance.

General Information	
Disabled	☐ Disable this server Set this option to disable this server without removing it from the list.
Server mode	Remote Access (User Auth)
Backend for authentication	authentication ldap authentication ldaps Local Database
Protocol	UDP on IPv4 only
Device mode	tun - Layer 3 Tunnel Mode "tun" mode carries IPv4 and IPv6 (OSI layer 3) and is the most common and compatible mode across all platforms. "tap" mode is capable of carrying 802.3 (OSI Layer 2.)
Interface	WAN The interface or Virtual IP address where OpenVPN will receive client connections.
Local port	1194 The port used by OpenVPN to receive client connections.
Description	OPENVPN SITKA A description may be entered here for administrative reference (not parsed).

Dans le champ **Peer Certificate Authority** on sélectionne notre autorité de certification
 Dans le champ **Server certificate** on sélectionne notre certificat

Cryptographic Settings	
TLS Configuration	☒ Use a TLS Key A TLS key enhances security of an OpenVPN connection by requiring both parties to have a common key before a peer can perform a TLS handshake. This layer of HMAC authentication allows control channel packets without the proper key to be dropped, protecting the peers from attack or unauthorized connections. The TLS Key does not have any effect on tunnel data.
	☒ Automatically generate a TLS Key.
Peer Certificate Authority	CA-Sitka
Peer Certificate Revocation list	No Certificate Revocation Lists defined. One may be created here: System > Cert. Manager
OCSP Check	☐ Check client certificates with OCSP
Server certificate	https_sitka (Server: Yes, CA: CA-Sitka, In Use)
DH Parameter Length	2048 bit Diffie-Hellman (DH) parameter set used for key exchange. ⓘ
ECDH Curve	Use Default The Elliptic Curve to use for key exchange. The curve from the server certificate is used by default when the server uses an ECDSA certificate. Otherwise, secp384r1 is used as a fallback.
Data Encryption Negotiation	☒ Enable Data Encryption Negotiation This option allows OpenVPN clients and servers to negotiate a compatible set of acceptable cryptographic data encryption algorithms from those selected in the Data Encryption Algorithms list below. Disabling this feature is deprecated.
Data Encryption Algorithms	AES-192-CFB (192 bit key, 128 bit block) AES-192-CFB1 (192 bit key, 128 bit block) AES-192-CFB8 (192 bit key, 128 bit block) AES-192-GCM (192 bit key, 128 bit block) AES-192-OFB (192 bit key, 128 bit block) AES-256-CFB (256 bit key, 128 bit block) AES-256-CFB1 (256 bit key, 128 bit block) AES-256-CFB8 (256 bit key, 128 bit block) AES-256-GCM (256 bit key, 128 bit block) AES-256-OFB (256 bit key, 128 bit block) AES-256-GCM AES-128-GCM CHACHA20-POLY1305 AES-256-CBC Available Data Encryption Algorithms Click to add or remove an algorithm from the list The order of the selected Data Encryption Algorithms is respected by OpenVPN. ⓘ
Fallback Data Encryption Algorithm	AES-256-CBC (256 bit key, 128 bit block) The Fallback Data Encryption Algorithm used for data channel packets when communicating with clients that do not support data encryption algorithm negotiation. This algorithm is automatically included in the Data Encryption Algorithms list.
Auth digest algorithm	SHA256 (256-bit) The algorithm used to authenticate data channel packets, and control channel packets if a TLS Key is present. When an AEAD Encryption Algorithm mode is used, such as AES-GCM, this digest is used for the control channel only, not the data channel. The server and all clients must have the same setting. While SHA1 is the default for OpenVPN, this algorithm is insecure.
Hardware Crypto	No Hardware Crypto Acceleration
Certificate Depth	One (Client+Server) When a certificate-based client logs in, do not accept certificates below this depth. Useful for denying certificates made with intermediate CAs generated from the same CA as the server.

Maintenant on va configurer le tunnel VPN.

- **IPv4 Tunnel Network** : En se connectant en VPN le client obtiendra une adresse IP dans ce réseau
- **IPv4 Local network** : les LAN qu'on veut rendre accessibles via le tunnel VPN. Dans notre cas, on va rendre accessible les réseaux 192.168.1.0/24, 192.168.2.0/24, 172.20.0.0/24
- **Concurrent connections** : le nombre de connexions VPN simultanés autorisés.

Tunnel Settings	
IPv4 Tunnel Network	192.168.1.0/24 This is the IPv4 virtual network used for private communications between this server and client hosts expressed using CIDR notation (e.g. 10.0.8.0/24). The first usable address in the network will be assigned to the server virtual interface. The remaining usable addresses will be assigned to connecting clients.
IPv6 Tunnel Network	This is the IPv6 virtual network used for private communications between this server and client hosts expressed using CIDR notation (e.g. fe80::/64). The ::1 address in the network will be assigned to the server virtual interface. The remaining addresses will be assigned to connecting clients.
Redirect IPv4 Gateway	☐ Force all client-generated IPv4 traffic through the tunnel.
Redirect IPv6 Gateway	☐ Force all client-generated IPv6 traffic through the tunnel.
IPv4 Local network(s)	172.20.0.0/24,192.168.1.0/24,192.168.2.0/24 IPv4 networks that will be accessible from the remote endpoint. Expressed as a comma-separated list of one or more CIDR ranges. This may be left blank if not adding a route to the local network through this tunnel on the remote machine. This is generally set to the LAN network.
IPv6 Local network(s)	IPv6 networks that will be accessible from the remote endpoint. Expressed as a comma-separated list of one or more IP/PREFIX. This may be left blank if not adding a route to the local network through this tunnel on the remote machine. This is generally set to the LAN network.
Concurrent connections	37 Specify the maximum number of clients allowed to concurrently connect to this server.
Allow Compression	Refuse any non-stub compression (Most secure) Allow compression to be used with this VPN instance. Compression can potentially increase throughput but may allow an attacker to extract secrets if they can control compressed plaintext traversing the VPN (e.g. HTTP). Before enabling compression, consult information about the VORACLE, CRIME, TIME, and BREACH attacks against TLS to decide if the use case for this specific VPN is vulnerable to attack. Asymmetric compression allows an easier transition when connecting with older peers.
Push Compression	☐ Push the selected Compression setting to connecting clients.
Type-of-Service	☐ Set the TOS IP header value of tunnel packets to match the encapsulated packet value.
Inter-client communication	☐ Allow communication between clients connected to this server
Duplicate Connection	☐ Allow multiple concurrent connections from the same user When set, the same user may connect multiple times. When unset, a new connection from a user will disconnect the previous session. Users are identified by their username or certificate properties, depending on the VPN configuration. This practice is discouraged security reasons, but may be necessary in some environments.

Pour les paramètres des clients, on coche **Dynamic IP** : Ceci aux client de garder leur connexion si l'adresse IP publique d'un client change, en se connectant via une connexion 4G .

Au niveau de la **Topology** , pour des raisons de sécurité, il vaut mieux utiliser la topologie **net30 - isolated /30 network per client** pour que chaque client soit isolé dans un sous-réseau de la plage réseau VPN afin que les clients ne puissent pas communiquer entre eux

Inconvénient de cette méthode la connexion VPN, va consommer 4 adresses IP : une adresse IP pour le PC, une adresse IP pour le pare-feu et les adresses de réseau et broadcast du sous-réseau en /30.

Client Settings	
Dynamic IP	☒ Allow connected clients to retain their connections if their IP address changes.
Topology	Subnet - One IP address per client in a common subnet Specifies the method used to supply a virtual adapter IP address to clients when using TUN mode on IPv4. Some clients may require this be set to "subnet" even for IPv6, such as OpenVPN Connect (iOS/Android). Older versions of OpenVPN (before 2.0.9) or clients such as Yealink phones may require "net30".

On coche l'option **Provide a DNS server list to clients**.

On indique le nom de domaine

On coche l'option "**Provide a default domain name to clients**" pour indiquer le nom de domaine local.

On indique les adresses IP de nos serveurs DNS

Aussi en force la mise à jour du cache dns

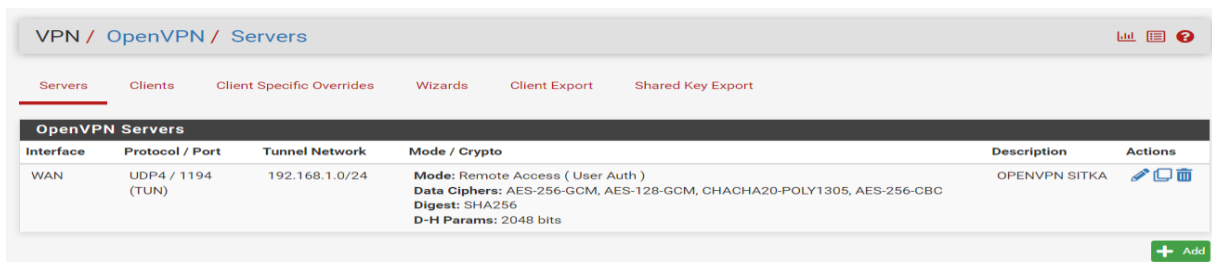
Advanced Client Settings	
DNS Default Domain	☒ Provide a default domain name to clients
DNS Default Domain	sitka.local
DNS Server enable	☒ Provide a DNS server list to clients. Addresses may be IPv4 or IPv6.
DNS Server 1	192.168.1.1
DNS Server 2	172.20.0.14
DNS Server 3	
DNS Server 4	
Block Outside DNS	☐ Make Windows 10 Clients Block access to DNS servers except across OpenVPN while connected, forcing clients to use only VPN DNS servers. Requires Windows 10 and OpenVPN 2.3.9 or later. Only Windows 10 is prone to DNS leakage in this way, other clients will ignore the option as they are not affected.
Force DNS cache update	☒ Run "net stop dnscache", "net start dnscache", "ipconfig /flushdns" and "ipconfig /registerdns" on connection initiation. This is known to kick Windows into recognizing pushed DNS servers.
NTP Server enable	☐ Provide an NTP server list to clients
NetBIOS enable	☐ Enable NetBIOS over TCP/IP If this option is not set, all NetBIOS-over-TCP/IP options (including WINS) will be disabled.

Dans la zone "**Custom options**", on tape : **auth-nocache**. Cette option refuse la mise en cache des identifiants

Advanced Configuration	
Custom options	auth-nocache Enter any additional options to add to the OpenVPN server configuration here, separated by semicolon. EXAMPLE: push "route 10.0.0.0 255.255.255.0"
Username as Common Name	☐ Use the authenticated client username instead of the certificate common name (CN). When a user authenticates, if this option is enabled then the username of the client will be used in place of the certificate common name for purposes such as determining Client Specific Overrides.
UDP Fast I/O	☐ Use fast I/O operations with UDP writes to tun/tap. Experimental. Optimizes the packet write event loop, improving CPU efficiency by 5% to 10%. Not compatible with all platforms, and not compatible with OpenVPN bandwidth limiting.
Exit Notify	Reconnect to this server / Retry once Send an explicit exit notification to connected clients/peers when restarting or shutting down, so they may immediately disconnect rather than waiting for a timeout. In SSL/TLS Server modes, clients may be directed to reconnect or use the next server. In Peer-to-Peer Shared Key or with a /30 Tunnel Network, this value controls how many times this instance will attempt to send the exit notification.
Send/Receive Buffer	Default Configure a Send and Receive Buffer size for OpenVPN. The default buffer size can be too small in many cases, depending on hardware and network uplink speeds. Finding the best buffer size can take some experimentation. To test the best value for a site, start at 512KiB and test higher and lower values.
Gateway creation	☒ Both ☐ IPv4 only ☐ IPv6 only If you assign a virtual interface to this OpenVPN server, this setting controls which gateway types will be created. The default setting is 'both'.
Verbosity level	3 (recommended) Each level shows all info from the previous levels. Level 3 is recommended for a good summary of what's happening without being swamped by output. None: Only fatal errors Default through 4: Normal usage range 5: Output R and W characters to the console for each packet read and write. Uppercase is used for TCP/UDP packets and lowercase is used for TUN/TAP packets. 6-11: Debug info range

Save

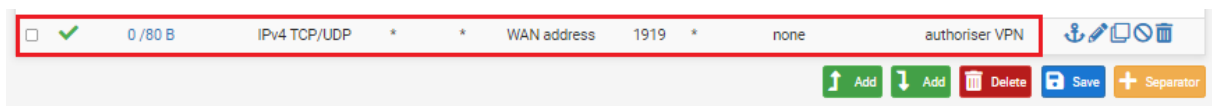
Maintenant on enregistre pour valider la configuration de notre VPN



3- Création des règles de firewall pour OpenVPN

a. Autoriser OpenVPN sur l'interface Wan

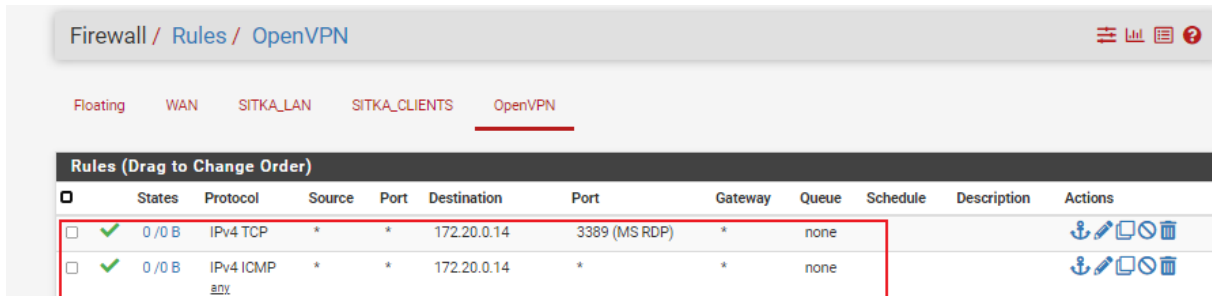
Il ne faut pas oublier de créer une règle autorisant le VPN sur notre interface wan



b. Autoriser l'accès aux ressources

Maintenant on va ajouter une règle, sur l'interface OpenVPN.

La règle va autoriser l'accès en RDP à l'hôte 172.20.0.14 au travers du tunnel VPN. On doit créer une ou plusieurs règles en fonction des ressources auxquelles nous voulons y accéder via le VPN.

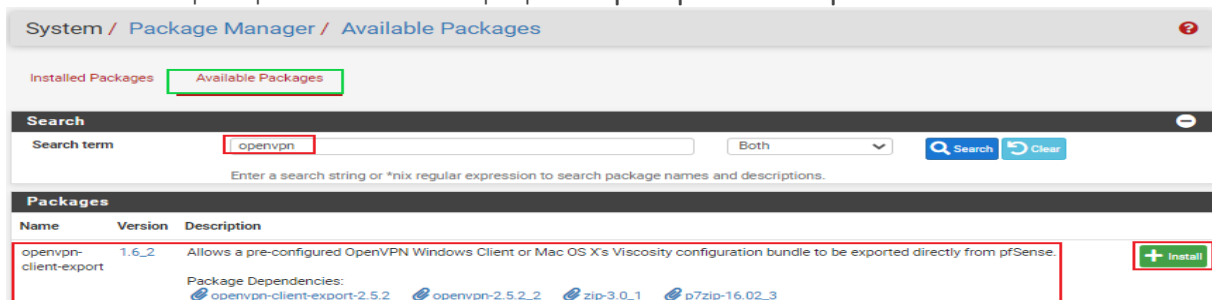


4- Exportation de la configuration OpenVPN

Pour exporter la configuration de notre openvpn pour les clients on va installer un paquet dont le nom est **openvpn-client-export** sur pfsense..

On va dans le menu suivant : **System > Package Manager > Available Packages**.

Recherchez "openvpn" et installez le paquet : **openvpn-client-export**.



Une fois l'installation est faite, on retourne dans le menu **OpenVPN** puis dans l'onglet **Client Export**.

Dans host name on rentre notre adresse IP publique pour nous connecter

OpenVPN / Client Export Utility

Server Client Client Specific Overrides Wizards **Client Export** Shared Key Export

OpenVPN Server

Remote Access Server: OPENVPN SITKA UDP4:1919

Client Connection Behavior

Host Name Resolution: Other

Host Name: 93.124.120.13
Enter the hostname or IP address the client will use to connect to this server.

Verify Server CN: Automatic - Use verify-x509-name where possible
Optionally verify the server certificate Common Name (CN) when the client connects.

Block Outside DNS: ☐ Block access to DNS servers except across OpenVPN while connected, forcing clients to use only VPN DNS servers. Requires Windows 10 and OpenVPN 2.3.9 or later. Only Windows 10 is prone to DNS leakage in this way, other clients will ignore the option as they are not affected.

Legacy Client: ☐ Do not include OpenVPN 2.5 settings in the client configuration. When using an older client (OpenVPN 2.4.x), check this option to prevent the exporter from placing known-incompatible settings into the client configuration.

Silent Installer: ☐ Create Windows installer for unattended deploy. Create a silent Windows Installer for unattended deployment. Since this installer is not signed, you may need special software to deploy it correctly.

Use Random Local Port: ☐ Use a random local source port (lport) for traffic from the client. Without this set, two clients may not run concurrently.

On tous les autres options par défaut, on rajoute seulement l'option **auth-nocache** ; après enregistre

Advanced

Additional configuration options: auth-nocache

Enter any additional options to add to the OpenVPN client export configuration here, separated by a line break or semicolon.

EXAMPLE: remote-random;

Save as default

5- Test de la connexion VPN depuis un poste client

a- Téléchargement du client openvpn

Selon le client on télécharge le programme adéquate

User	Certificate Name	Export
Authentication Only (No Cert)	none	Inline Configurations: Most Clients Android OpenVPN Connect (iOS/Android) Bundled Configurations: Archive Config File Only Current Windows Installers (2.5.2-1x01): 64-bit 32-bit Legacy Windows installers (2.4.11-1x01): 10/2016/2019 7/8/8.1/2012r2 Viscosity (Mac OS X and Windows): Viscosity Bundle Viscosity Inline Config

Only OpenVPN-compatible user certificates are shown

On teste l'accès distant depuis un poste client Windows, Android ou Apple . Pour apple on choisit Viscosity gratuit un mois après c'est payant ou Tunneblick qui est gratuit

If a client is missing from the list it is likely due to a CA mismatch between the OpenVPN server instance and the client certificate, the client certificate does not exist on this firewall, or a user certificate is not associated with a user when local database authentication is enabled.

OpenVPN 2.4.8+ requires Windows 7 or later

Links to OpenVPN clients for various platforms:

OpenVPN Community Client - Binaries for Windows, Source for other platforms. Packaged above in the Windows Installers

OpenVPN For Android - Recommended client for Android

FEAT VPN For Android - For older versions of Android

OpenVPN Connect: [Android \(Google Play\)](#) or [iOS \(App Store\)](#) - Recommended client for iOS

Viscosity - Recommended commercial client for Mac OS X and Windows

Tunnelblick - Free client for OS X

Using the Latest OpenVPN on Linux Distro - Install OpenVPN using the OpenVPN apt repositories to get the latest version, rather than one included with distributions.

b- Test de connexion sur un client Windows

Sur un PC client, on installe le client OpenVPN ⁷³ ~~ss~~ openvpn-heimdall-UDP4-1919-install-2.5.2-I601-amd64

Setup OpenVPN 2.5.2-I601

Choose setup type.

Install Now

Customize

Setup OpenVPN 2.5.2-I601

Installing OpenVPN...

Deleting TUN/TAP adapters

Setup OpenVPN 2.5.2-I601

OpenVPN Installing Completed

Close

OpenVPN Configuration Setup

Welcome to OpenVPN installer.

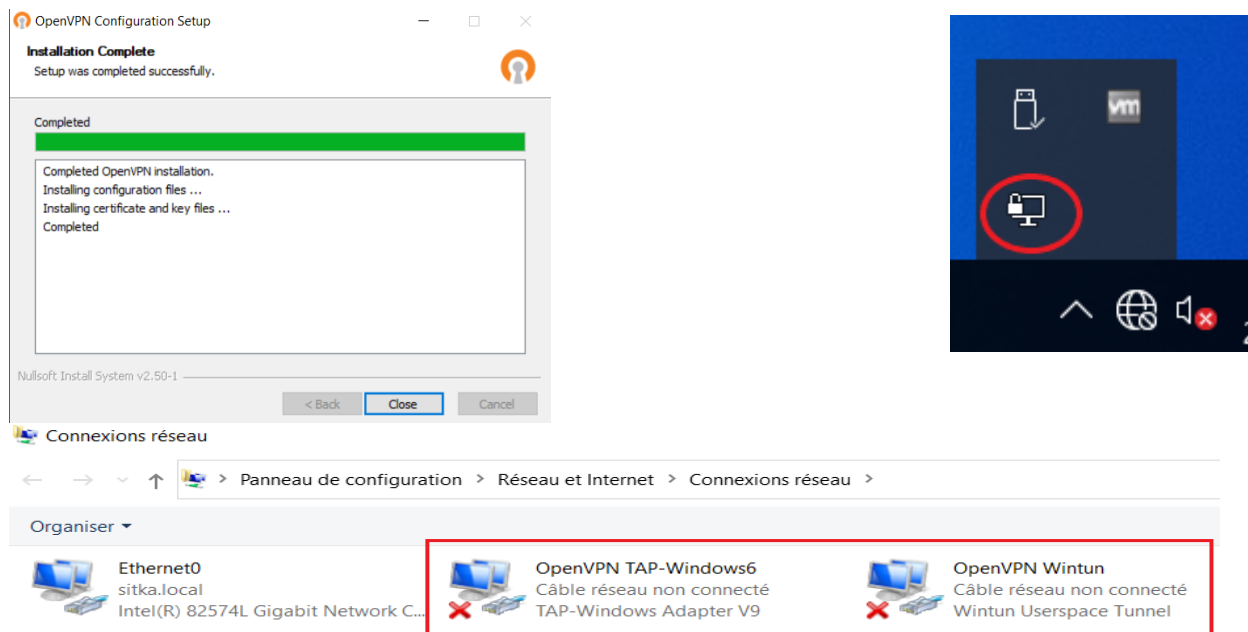
This wizard will guide you through the installation of the OpenVPN client and configuration.

This will automatically install the configuration files needed for your connection.

Install

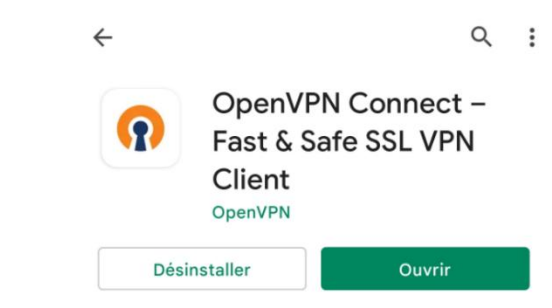
Cancel

Une fois l'agent VPN est installé un icône dans la barre des tâches apparait qui va ne permettre de se connecter à notre serveur VPN ainsi que les interfaces réseaux propre au VPN comme c'est illustré ci-dessous



c- Test de connexion sur un client Android

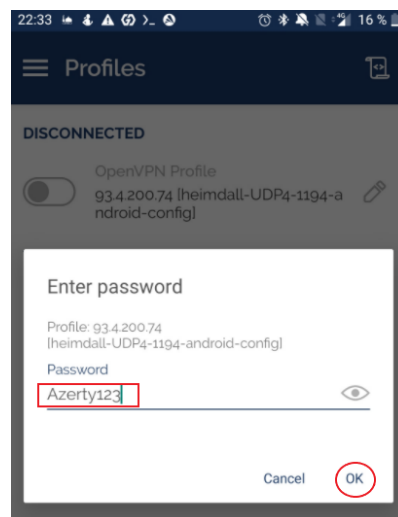
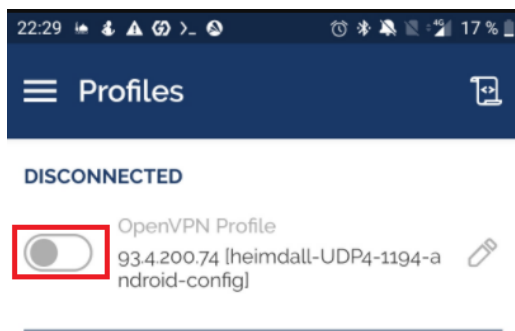
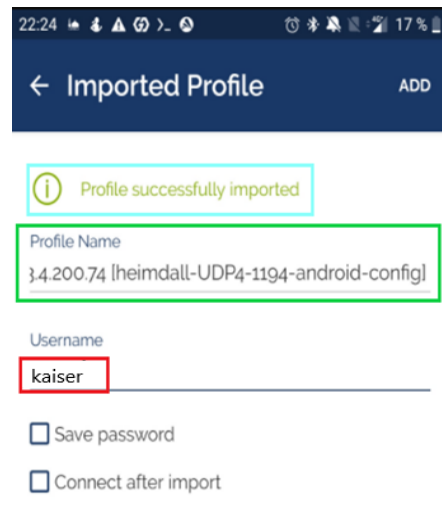
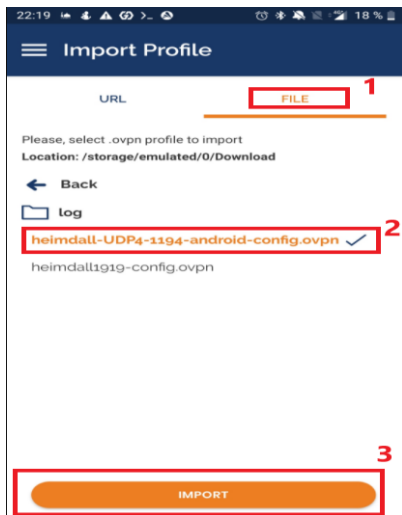
Sur google store on télécharge l'application openvpn



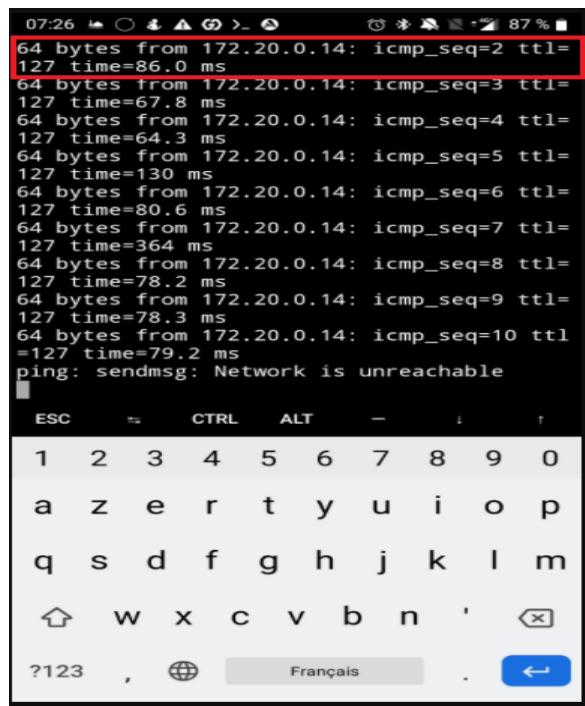
Sur l'interface web de pfsense on télécharge le client androïde pour smartphone en cliquant

Sur 

Une fois Le fichier que le fichier est téléchargé [heimdall-UDP4-1194-android-config](#) on l'envoi par mail et on le récupère sur notre messagerie sur le smartphone, Ce fichier on va l'enregistrer et on va l'injecter dans notre application openvpn Android, après on rentre notre identifiant utilisateur et notre mot de passe



Voila la connexion est établie de mon smartphone connecté en 4G sur mon serveur Pfsense connecté sur la box Internet de mon fournisseur d'accès ; je peux vérifier mon accès à hermes avec un ping depuis mon smartphone



De même Sur l'interface web (Status+Openvpn] la connexion vpn du client Android est enregistré

